
[TODO: working title – suggested: “Merging Specialist Diffusion Models for Multi-Task Astronomical Image Restoration”]

July 9, 2026

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Abstract

Astronomical image-processing pipelines typically chain several independent restoration steps – e.g. star removal, defect correction, super-resolution – each best solved by a dedicated model. We investigate whether such a pipeline can be collapsed into a *single* set of weights via post-hoc model merging, avoiding both the cost of serving multiple networks and the need for joint multi-task training data. Three v-prediction DDPMs are independently fine-tuned – one per task (star removal, dead/hot-pixel restoration, super-resolution) – from a common, purpose-pretrained initialization θ_0 , and their task vectors are combined with Task Arithmetic, TIES-Merging, and DARE-TIES. Crucially, because initial zero-shot merging configurations failed to meet performance expectations due to destructive weight interference, we introduce a brief post-merge fine-tuning stage for all three methods. Merging and fine-tuning hyper-parameters are chosen via a two-stage grid search evaluated on all seven combinations of the three degradations, using real, manually-curated Hubble Legacy Archive (HLA) imagery. **[TODO: one-sentence headline result once the grid search is finalized].**

1. Introduction

Restoring astronomical images typically requires several corrections applied in sequence: removing point-like sources (stars) to reveal extended low-surface-brightness structure, correcting detector defects (dead/hot pixels), and recovering high-frequency detail lost to instrument or seeing-limited resolution (super-resolution). A natural deep-learning approach is to train one specialist model per

correction; the drawback is that deploying the pipeline still requires keeping and running several networks, and any joint model would require a dataset with all degradations co-occurring and annotated.

Model merging – combining independently fine-tuned checkpoints directly in weight space – has recently been shown to recover much of the performance of joint multi-task training for classification and language models (Ilharco et al., 2023; Yadav et al., 2023) with little to no additional training. However, we find that applying these methods zero-shot to conditional diffusion models solving distinct low-level image restoration tasks fails to meet basic performance expectations. To overcome this limitation, we extend the standard merging paradigm by introducing a mandatory, lightweight post-merge fine-tuning phase for all evaluated methods, which successfully resolves interference in the merged weight space.

Concretely, we (i) pretrain a single shared initialization θ_0 on a generic conditional-restoration pretext task built from all our data, (ii) independently fine-tune three specialist DDPMs from θ_0 , one per degradation, (iii) merge the resulting task vectors with Task Arithmetic (Ilharco et al., 2023), TIES-Merging (Yadav et al., 2023), and DARE-TIES (Yu et al., 2024), and (iv) subject all merged models to a brief post-merge fine-tuning step to recover target performance, optimizing coefficients via a two-stage grid search.

[TODO: 1–2 sentences summarizing contributions as a list, once results are final, e.g. “(1) we show that X; (2) we find that TIES/DARE-TIES combined with post-merge fine-tuning outperforms naive Task Arithmetic by Y%; (3) we release code at <https://github.com/<todo>>.”]

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Code. **[TODO: insert GitHub link once the repository is finalized (submission requires the code to be reachable even if not public).]**

2. Related Work

Diffusion models. Denoising Diffusion Probabilistic Models (DDPMs) learn to invert a fixed noising process (Ho et al., 2020). We adopt the *v-prediction* parameterization (Salimans & Ho, 2022), a cosine noise schedule (Nichol & Dhariwal, 2021), and classifier-free-style condition dropout (Ho & Salimans, 2022) so that the network retains some unconditional generative capacity; the backbone is a conditional U-Net (Ronneberger et al., 2015) with self-attention (Vaswani et al., 2017) at low resolutions.

Model merging. Task Arithmetic (Ilharco et al., 2023) showed that the difference between a fine-tuned and a pre-trained checkpoint (“task vector”) can be added back to the pretrained weights, with a scalar coefficient λ , to transfer or combine capabilities without further training. TIES-Merging (Yadav et al., 2023) improves on naive summation when merging several task vectors by trimming low-magnitude coordinates and resolving sign disagreements before averaging, reducing destructive interference. DARE (Yu et al., 2024) randomly drops and rescales a large fraction of each task vector’s coordinates prior to merging, and is commonly composed with TIES (“DARE-TIES”). All three methods require every fine-tuned checkpoint to originate from the *same* pretrained weights, which directly motivates our shared- θ_0 pretraining stage (Sec. 3.2).

Model merging has been studied predominantly for classifiers and LLMs; in the diffusion setting it has mostly been applied to merging LoRA adapters for text-to-image personalization rather than to merging fully fine-tuned specialist denoisers trained on distinct low-level restoration tasks. Crucially, while literature treats model merging as a zero-shot operations, our findings show that zero-shot merging does not meet performance expectations for pixel-level generative corrections, thus justifying our exploration of post-merge fine-tuning. **[TODO: verify/extend this claim with a short literature search closer to submission – this is the part of the related work most likely to be out of date, and it is also our best candidate for a genuine novelty claim, so it deserves a careful check rather than a hedge.]**

Star removal. We use StarNet (Misiura, 2020), a pre-trained convolutional encoder–decoder that removes stars from an astronomical image in a single pass, only to *generate training pairs* for our specialist (Sec. 3.1) – not as a teacher network in a distillation sense.

3. Method

3.1. Data and per-task degradation pipeline

Source imagery. We start from a manually curated set of color, already-drizzled mosaics from the Hubble Legacy Archive (HLA) (Space Telescope Science Institute (STScI), 2006). **[TODO: how many source frames, which instruments/proposals, native resolution range, selection criteria.]** Raw frames are converted to a working dynamic range via a nonlinear stretch before any degradation is applied. **[TODO: exact stretching function and parameters (e.g. black point, softening parameter, per-channel vs. luminance-only, output range); if this is the same arcsinh stretch used elsewhere in the project, say so explicitly and cite/describe it here.]**

For all three tasks the supervision target is always the fully clean, stretched image; only the *condition* (network input) differs. Each specialist therefore sees, as condition, an image degraded with only its own corruption, but is only ever supervised to recover the clean image – as opposed to θ_0 ’s pretext task (Sec. 3.2), where the degradation is deliberately *not* tied to the task the source image came from.

1. **Star removal and synthetic injection.** The clean image is passed through StarNetV2 (Misiura, 2020) to obtain a preliminary starless prediction. Rather than using this output directly as a ground-truth target, which would reduce our model to a mere distillation of StarNet, we use it solely to isolate real stars. A library of isolated real star patches (typically 64×64 pixels) is built by subtracting the StarNet prediction from the original image and extracting regions around local intensity maxima. These extracted stars are then synthetically re-injected onto pristine starless backgrounds via linear blending to form strictly controlled (*synthetic-starry, clean*) training pairs. Crucially, applying a uniform, global stretching function across the entire dataset yielded highly suboptimal results; due to the vast diversity in exposure levels, noise floors, and dynamic ranges among different astronomical targets, a static stretch caused severe clipping or contrast degradation in many samples. To overcome this, an image-specific adaptive arcsinh stretch is performed on each pair before training. As established in our preprocessing pipeline, these personalized stretch parameters (specifically the 1st and 99.7th percentiles for black and white points) are computed exclusively on the primary starry image. These exact parameters are then applied rigidly to the paired starless target, ensuring flawless photometric consistency, preventing artificial background shifts, and preserving the true underlying gradient.

2. **Dead/hot-pixel restoration.** These targets are gener-

ated using a robust outlier-rejection pipeline that isolates anomalies by evaluating the residual between each pixel and a local median filter (typically using a 5×5 window). A pixel is flagged as defective if its channel residual exceeds a threshold of 6.0σ , where σ is estimated via a robust Median Absolute Deviation (MAD) scaled by a factor of 1.4826.

Because the HLA imagery consists of RGB composites generated from independent, non-simultaneous single-filter exposures, these defects exhibit a sparse spatial distribution and strict channel independence (i.e., a hot pixel often corrupts only a single color channel). In terms of saturation behavior, hot pixels present high-intensity positive spikes approaching clipping thresholds (1.0), whereas dead pixels manifest as sharp drops toward the local noise floor (0.0). During training, a randomized patch sampling strategy extracts sub-images exclusively from regions verified by a geometric validity mask; this effectively filters out empty edge-padding zones while preserving the natural, sparse defect rate (typically under 0.05%) for implicit network regularization.

3. **Super-resolution.** To condition the generative model for image enhancement, a paired dataset is constructed by applying a classical degradation pipeline to the high-resolution (HR) ground-truth images. Each HR image is first smoothed using a spatial Gaussian blur kernel ($\sigma = 1.5$) to simulate Point Spread Function (PSF) broadening and telescope optical limitations. The blurred image is then downsampled by a factor of $\times 4$ using a third-order cubic spline interpolation to drastically reduce structural information. Crucially, to inject this low-resolution input as a conditioning channel into the Denoising Diffusion Probabilistic Model (DDPM), the degraded image is upsampled back to the original HR spatial dimensions via bicubic interpolation, ensuring structural alignment with the target. During training, synchronized spatial patches are cropped on-the-fly from these full-scale pairs to feed the network while preserving exact coordinate matching between the degraded conditioning map and the HR target.

[TODO: Table with per-task dataset sizes (raw source images, extracted patches) and train/val/test split fractions – see placeholder Table 1.]

3.2. Shared initialization θ_0

All three specialists are fine-tuned from a common checkpoint θ_0 , obtained by pretraining the same architecture from a random initialization on a *generic conditional-restoration* pretext task, rather than starting each special-

Table 1. Dataset composition per specialist task. **[TODO: fill in raw numbers.]**

Task	Source imgs.	Patches	Train	Val/Test
Star removal (SR)	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
Restoration (IR)	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
Super-res. (SU)	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]

FIGURE PLACEHOLDER
 pipeline diagram: raw HLA frame $\rightarrow$ stretch $\rightarrow$ per-task degradation $\rightarrow$ (condition, target) pair, one row per task

Figure 1. **[TODO: Overview of the data pipeline from raw HLA mosaics to the three per-task (condition, target) datasets.]**

ist from an independent random init. This choice is required by the merging methods themselves: Task Arithmetic, TIES-Merging and DARE-TIES all define each specialist’s contribution as a task vector $\tau_t = \theta_t - \theta_0$, which is only meaningful if every θ_t was fine-tuned from the *same* origin θ_0 (Ilharco et al., 2023; Yadav et al., 2023). Beyond compatibility, we hypothesize that a shared low-level restoration prior (rather than random weights) also makes fine-tuning more sample-efficient and yields specialists that are better aligned in weight space, which prior work links to lower interference when merging (Yadav et al., 2023). **[TODO: this is currently a hypothesis, not validated: consider an ablation comparing merging performance of θ_0 -initialized specialists vs. three independently random-initialized specialists.]**

Architecture. The core generative model relies on a conditional U-Net architecture (*ImprovedUNet*) that processes a 6-channel tensor obtained by concatenating the noisy latent x_t ($C = 3$) and the spatial conditioning map ($C = 3$). The network features an initial feature extraction block followed by 4 downsampling and 4 upsampling stages. Downsampling is performed via strided convolutions (4×4 , stride 2), which progressively double the channel width at each stage ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$).

Each architectural stage is built around residual *UNet-Blocks* comprising two convolutional layers (3×3), Group Normalization (4 groups), and SiLU activations. Timestep embedding is achieved using a Sinusoidal projection layer ($C = 64$) followed by a two-layer Multi-Layer Perceptron (MLP) with GELU activation to output a 256-dimensional time embedding vector, which is linearly projected and added to the intermediate features of every residual block.

Multi-head self-attention mechanisms (Vaswani et al., 2017) (configured with 2 heads and Layer Normalization) are integrated at the highest abstraction levels: specifically, at the end of the third encoder stage ($C = 512$), within the central bottleneck layer ($C = 512$), and immediately after the first decoder block ($C = 256$). The decoder utilizes transposed convolutions (4×4 , stride 2) for spatial up-scaling, followed by standard channel-wise concatenation with the encoder’s skip connections. This results in concatenated channel dimensions of 1024, 512, 256, and 128 before being reduced by their respective decoder blocks. A final convolutional layer projects the remaining 64 channels back to the 3-channel image space to predict the v -target. This configuration remains strictly invariant across θ_0 pre-training and all downstream fine-tuning tasks.

Diffusion formulation. We use v -prediction (Salimans & Ho, 2022) with a cosine noise schedule (Nichol & Dhariwal, 2021) (200 diffusion steps at pretraining time), optimized with the standard denoising MSE loss between predicted and true v .

Pretext task. θ_0 is trained to map a *synthetically* degraded image back to its clean version. Targets are the pooled union of the training-split clean images from all three downstream tasks (val/test splits are never touched at this stage). Conditions are obtained by applying 1–3 randomly chosen operations from {additive Gaussian/Poisson-like noise, Gaussian blur, bicubic downsample-upsample, small random occlusions filled with local-mean statistics}, with severity/parameters drawn independently of which task the source image came from, and a 10% chance of a fully zeroed condition (classifier-free-guidance-style dropout (Ho & Salimans, 2022)). This deliberately decorrelates image content from degradation type, so that θ_0 does not learn any implicit content $\leftrightarrow$ corruption association that a specialist fine-tune would then have to unlearn.

3.3. Specialist fine-tuning

Each specialist reuses θ_0 ’s architecture (an Improved U-Net with 64 base channels) and the v -prediction framework, incorporating a cosine noise schedule over $T = 200$ diffusion steps, an exponential moving average (EMA) with $\beta = 0.999$, and the AdamW optimizer. Fine-tuning is conducted exclusively on each specialist’s respective (con-

dition, target) pairs from Sec. 3.1. The optimization utilizes a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} , regulated by a cosine annealing learning rate scheduler over a total budget of 1000 epochs (contracted to a clean stop at 100 epochs per session for resource management). Training is distributed via Hugging Face Accelerate across 2 GPUs with a local batch size of 4 per device, yielding a global batch size of 8. A SHA-256 hash check on θ_0 is used across the fine-tuning runs to guarantee all specialists literally start from the same weights.

3.4. Model merging and post-merge fine-tuning

Given task vectors $\tau_t = \theta_t - \theta_0$ for $t \in \{\text{SR, IR, SU}\}$, we first construct combined weight configurations using:

- **Task Arithmetic** (Ilharco et al., 2023): $\theta_{\text{merged}} = \theta_0 + \lambda \sum_t \tau_t$.
- **TIES-Merging** (Yadav et al., 2023): each τ_t is trimmed to its top- k magnitude coordinates, a majority sign is elected per coordinate across the trimmed vectors, and only coordinates agreeing with the elected sign are averaged, before scaling by λ .
- **DARE-TIES** (Yu et al., 2024): each τ_t is first sparsified via DARE’s random drop-and-rescale before applying TIES.

Post-merge fine-tuning. In early experiments, we noticed that zero-shot application of Task Arithmetic, TIES, and DARE-TIES failed to meet baseline performance expectations, resulting in significant image artifacts and a severe drop in reconstruction fidelity due to destructive weight interference. To address this, we introduce a subsequent, mandatory fine-tuning stage for all three merged checkpoints. This phase consists of a brief training run utilizing a low learning rate and an aggregated dataset containing balanced mixtures of all three tasks, smoothly restoring cross-task synergy and structural regularities.

Two-stage grid search. [TODO: this section is the least complete and will be updated once the grid-search notebook is finalized. Currently missing: the exact search ranges for λ (shared vs. one per specialist), TIES trim density k , DARE drop rate p ; the reconstruction/perceptual metric(s) used to score a configuration; the exact “retention” formula (score of the merged model relative to the corresponding single-task specialist, expressed as a %) and how it is aggregated; the Stage-1 filtering threshold; and the number of configurations evaluated / compute budget.] At a high level: Stage 1 scores candidate merged configurations after a fixed number of post-merge fine-tuning iterations on the six

[TODO: or seven?] single-degradation evaluation folders and discards clearly underperforming ones; Stage 2 re-evaluates the surviving configurations on all combinations of degradations (up to and including all three simultaneously) to pick the final merging and fine-tuning recipe.

4. Experimental Setup

Evaluation data. All seven combinations of the three degradations (SR, IR, SU, and their pairwise/triple compositions) are constructed from a held-out set of clean images, with identical crop positions across all seven folders (via a shared crop manifest) so that scores are directly comparable. **[TODO: table with per-folder sizes.]**

Baselines. **[TODO: confirm baselines used, e.g. (a) the three specialists run sequentially / in cascade as an upper-bound reference; (b) unweighted averaging ($\lambda_t = 1/3$ each, no trimming/sign-election) as a merging lower bound.]**

Metrics. **[TODO: list metrics (e.g. PSNR, SSIM, LPIPS, and/or task-specific scores) and whether they are computed against the clean target or against the corresponding single-task specialist’s own output.]**

Hardware / compute. **[TODO: GPU type/count, total training, post-merge fine-tuning, and grid-search compute budget.]**

5. Results

5.1. Task Arithmetic **[TODO: "error tracking pipeline"?)**

This section contains all the experiments performed initially on task arithmetic, and subsequently repliacted on the other methods directly on the last iteration in other to maximize the results. **[TODO: write better]**

Raw, pooled MSE. The earliest attempts scored candidate merges by MSE against ground truth, first at a single hand-picked λ point and then over an actual grid, split into a two-stage. However both consistently picked $\lambda_{SU} = 0$: because the super-resolution folder carries by far the largest absolute MSE, any pooled objective is minimised by *suppressing* that task rather than balancing it.

Normalised MSE. We proceeded to measure each task’s error relative to its own single-task expert,

$$\text{score} = \frac{1}{|T|} \sum_t \frac{\text{MSE}_{\text{merge}}[t]}{\text{MSE}_{\text{expert}}[t]},$$

alongside PSNR/SSIM for a fuller picture. This normalised objective was the first to select a genuine three-task compromise instead of switching a task off entirely. Combined-degradation folders have no single expert to normalise against, so they were instead scored relative to θ_0 — i.e. how much merging improves on doing nothing.

Perceptual Z-score. The final refinement replaced MSE with LPIPS, a perceptual metric, in order to compensate any mismatch between low MSE and visibly poor super-resolution output quality. More specifically, LPIPS was used as a replacement for MSE on any folder involving super-resolution, and put every folder on a common scale via a per-folder Z-score before ranking — $z = (x - \mu)/\sigma$ across the candidate grid, averaged before ranking.

[TODO: in sta tabella vanno acchittati i numeri perché non va bene + centra le robe]

Selection metric	λ grid	Winner
Raw MSE, single point	1 pt	0.9/0.1/0.0
Raw MSE, two-stage grid	27 combos	1.0/0.1/0.0
Normalised MSE (proposed)	≤ 4 combos	truncated run
Normalised MSE, two-stage, bug fixed	36 combos	0.9/0.1/0.1
Perceptual Z-score (LPIPS + MSE)	8 combos	0.9/0.1/0.1

Table 2. Evolution of the search objective. Values of λ reported as (SR/IR/SU).

What the diagnostics actually showed Even though the last two objectives agree on 0.9/0.1/0.1, per-task diagnostics on the super-resolution folder show this merge is not the clean win it appears to be:

Model	Det. val loss	MSE	LPIPS
Pure pre-trained base θ_0	0.0835	0.00827	0.2303
Single-task super-resolution expert	0.0880	0.00893	0.2203
Merged model (0.9/0.1/0.1)	0.1557	0.00707	0.4080

Table 3. Super-resolution folder: deterministic val. loss, MSE, LPIPS.

The merge has the *lowest* MSE of the three, yet by far the *worst* LPIPS: MSE was rewarding a smoother, perceptually worse prediction. The LPIPS gap persists even with many more sampling steps, ruling out a sampling artifact. The task vectors are nearly orthogonal (cosine similarity ≈ 0.08 – 0.12), which normally favours task arithmetic, but τ_{SU} ’s much larger norm means even a small mixing coeffi-

cient from the other two tasks perturbs its direction enough to visibly degrade perceptual quality.

[TODO: non ho capito che cosa vuole dire il paragrafo prima di questa scritta]

Findings The whole arc reduces to one lesson: **the choice of evaluation metric mattered far more than the choice of search algorithm.** Raw pooled MSE quietly answered a different question — “minimise total pixel error” — than the one actually being asked — “be good at all three tasks at once” — which is why it kept switching super-resolution off. Normalising against per-task baselines fixed that pooling artifact and recovered a real compromise, but only a perceptual metric exposed the deeper issue: interference between task vectors that pixel-wise error cannot see. Any merge of these experts should be validated perceptually before being trusted; promising next steps are interference-aware merging (e.g. TIES/DARE), renormalising the super-resolution task vector, or building it from an earlier, smaller-norm checkpoint.

Experiment	Main objective	Outcome
Baseline merge	Standard TIES merging with default aggregation.	Merged model consistently underperformed both specialists and, in several tasks, the pre-trained model.
Diagnostic analysis	Determine whether the performance loss originated from the specialists or the merge process.	Specialists were validated; the merge procedure was identified as the primary bottleneck.
Improved merging	Evaluate alternative aggregation strategies and expanded hyperparameter search.	Clear improvement over the baseline merge, with better task retention and higher restoration quality.
Extended evaluation	Assess merges on unseen combinations of degradations using a two-stage selection procedure.	Evaluation framework completed, but final testing was interrupted by a generic pretraining error.

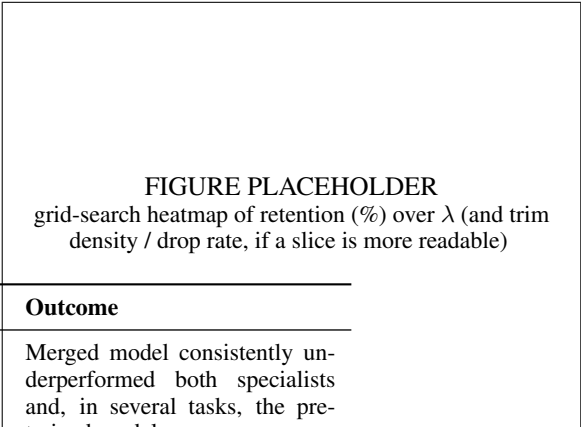
Table 4. Evolution of the TIES-merging experiments and their main outcomes.

6. Discussion and Conclusions

[TODO: Full section pending results. Points to address once numbers are available: (1) does any merging method approach the cascaded-specialists upper bound, and on which folders does it fall short (likely the multi-degradation ones, where interference is expected to be worst)? (2) Analysis of why zero-shot merging methods failed to satisfy expectations, and how much the proposed post-merge fine-tuning recovered multi-task performance; (3) does TIES/DARE-TIES meaningfully outperform plain Task Arithmetic after the fine-tuning phase, and is the improvement worth the

Table 5. Retention (%) of merged-model performance relative to the corresponding single-task specialist, per evaluation folder. [TODO: fill in.]

Eval. folder	Task Arith.	TIES	DARE-TIES
SR only	[TODO: -]	[TODO: -]	[TODO: -]
IR only	[TODO: -]	[TODO: -]	[TODO: -]
SU only	[TODO: -]	[TODO: -]	[TODO: -]
SR+IR	[TODO: -]	[TODO: -]	[TODO: -]
SR+SU	[TODO: -]	[TODO: -]	[TODO: -]
IR+SU	[TODO: -]	[TODO: -]	[TODO: -]
SR+IR+SU	[TODO: -]	[TODO: -]	[TODO: -]



[TODO: Grid search landscape for the best-performing merging method.]
extra hyperparameters (K, D)? (4) sensitivity of the result to λ – is there a wide plateau or a sharp optimum? (5) limitations: single, fairly small, HLA-specific datasets; dependence on a custom-pretrained backbone; Star-Net’s own biases propagating into the star-removal specialist through the synthetic star-injection pipeline. (6) future work: learned/optimized merging coefficients (e.g. AdaMerging-style test-time entropy minimization) instead of grid search; extending to more than three tasks; comparing against an explicitly multi-task-trained model as a further baseline.]

7. Statement on the use of AI

[TODO: This section is mandatory and must be accurate and specific – fill in honestly before submission. Template below; edit freely.]

AI tooling ([TODO: name the tool(s), e.g. “Claude”]) was used in this project for: [TODO: list concretely, e.g. “(i) debugging the pretraining/fine-tuning scripts (e.g. identifying that all three checkpoint paths pointed to an intermediate epoch_100 snapshot instead of the fi-

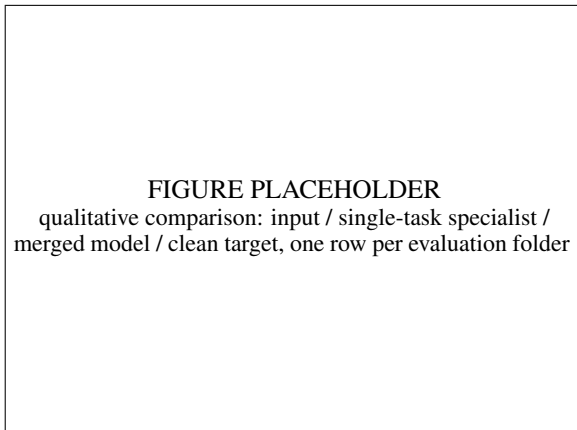


Figure 3. [TODO: Qualitative comparison across merging methods.]

nal epoch; that θ_0 hash verification had been silently dropped; that evaluation seeding was missing from the grid search); (ii) drafting boilerplate code for the crop-manifest / dataset-splitting scripts; (iii) editing and structuring this report; (iv) literature search for the Related Work section.”] All reported architectural choices, training curves, and quantitative results were produced and verified by the author; [TODO: any AI-suggested code or claim that ended up in the final pipeline should be explicitly flagged as such here, per the course’s AI-misuse policy.]

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Appendix

[TODO: Optional: move here anything cut from the main 2 pages, e.g. the full grid-search table, extra qualitative samples, the exact degradation-pipeline hyperparameters once finalized, or a derivation of the TIES trim/elect/merge steps if you want to show you understand them in detail.]